# Webasto Battery Solutions
# Series Sample End to End communication

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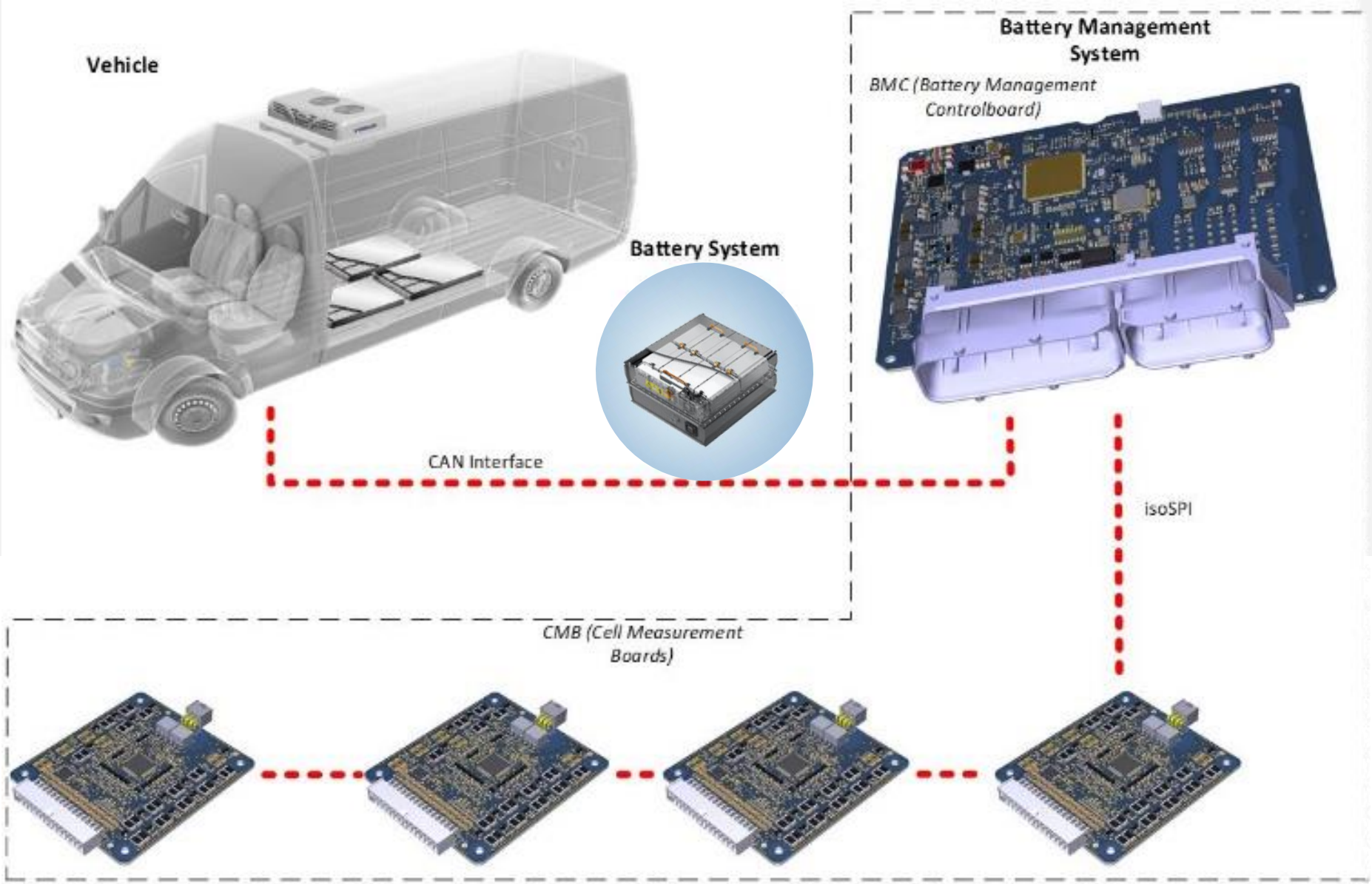
# Battery CAN communication

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# Software | End to End communication standard

Strictly Confidential
No Sharing

| Name | Start Position | Update Bit | Length [Bit] | Senders | Receivers |
|---|---|---|---|---|---|
| VehInfo | 0 | | 64 | Vehicle_Control_... | VIC |
| VehInfo_Cks | 0 | | 8 | Vehicle_Control_... | VIC |
| VehInfo_DIDNibble | 12 | | 4 | Vehicle_Control_... | VIC |
| TAmb | 16 | | 11 | Vehicle_Control_... | VIC |
| VehSpd | 32 | | 11 | Vehicle_Control_... | VIC |
| VehInfo_Alvctr | 8 | | 4 | Vehicle_Control_... | VIC |

## VehInfo – Message (Vehicle Information)

- The Battery needs every 100 ms following vehicle information (descried in ARXML- File):
  - VehInfo_Cks: Message Checksum (Calculated in Autosar Profile1C chapter 5.3.8 “Autosar_PRS_E2EProtocol.pdf”)
  - VehInfo_DIDNibble: Unique Data ID (Constant Value based on Autosar Profile1C (delivered by Webasto))
  - TAmb: Ambient Temperature [°C]
  - VehSpd: Vehicle Speed [km/h]
  - VehInfo_Alvctr: Message Alive Counter (incremented each telegram by 0x01, starts again after 0x0E → 0x00)

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- Battery needs every 100 ms “VehInfo”- message from the vehicle. -

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| Name | Start Po... | Update Bit | Length [Bit] | Senders | Receivers |
|---|---|---|---|---|---|
| CtlCmd | 0 | | 64 | Vehicle_Control_Unit | VIC |
| CtlCmd_Cks | 0 | | 8 | Vehicle_Control_Unit | VIC |
| CtlCmd_DIDNibble | 12 | | 4 | Vehicle_Control_Unit | VIC |
| HvReq | 16 | | 8 | Vehicle_Control_Unit | VIC |
| IslnMeastDisaCmd | 24 | | 2 | Vehicle_Control_Unit | VIC |
| CtlCmd_Alvctr | 8 | | 4 | Vehicle_Control_Unit | VIC |

## CtlCmd - Message (control command)

- The Battery needs every 100 ms following vehicle information (descried in ARXML- File):
  - CtlCmd_Cks: Message Checksum (Calculated in Autosar Profile1C chapter 5.3.8 “Autosar_PRS_E2EProtocol.pdf”)
  - CtlCmd_DIDNibble : Unique Data ID (Constant Value based on Autosar Profile1C (delivered by Webasto))
  - HvReq: High Voltage Request

    255: info not available (default)
    50: disconnect DC char. (VIB only)
    49: connect for DC char. (VIB only)
    48: prepare for DC char. (VIB only)
    32: connect for AC charging
    16: connect for driving
    1: emergency disconnect
    0: disconnect
  - IsInMeasDisaCmd: Disconnection of Isolation Monitoring

    3: info not available (default)
    2: reserved
    1: deactivate
    0: activate
  - CtlCmd_Alvctr: Message Alive Counter (incremented each telegram by 0x01, starts again after 0x0E → 0x00)

- Battery needs every 100 ms “CtlCmd”- message from the vehicle. -

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| Name | Start Po... | Update Bit | Length [Bit] | Senders | Receivers |
|---|---|---|---|---|---|
| SftyInfo | 0 | | 64 | Vehicle_Co... | VIC |
| SftyInfo_Cks | 0 | | 8 | Vehicle_Co... | VIC |
| SftyInfo_DIDNibble | 12 | | 4 | Vehicle_Co... | VIC |
| SftyOk | 16 | | 2 | Vehicle_Co... | VIC |
| CrashDetnNeg | 18 | | 2 | Vehicle_Co... | VIC |
| SftyInfo_Alvctr | 8 | | 4 | Vehicle_Co... | VIC |

## SftyInfo - Message (safety information from vehicle)

- The Battery needs every 10 ms following vehicle information (descried in ARXML- File):
  - SftyInfo_Cks: Message Checksum (Calculated in Autosar Profile1C chapter 5.3.8 “Autosar_PRS_E2EProtocol.pdf”)
  - SftyInfo_DIDNibble: Unique Data ID (Constant Value based on Autosar Profile1C (delivered by Webasto))
  - SftyOk: Status of Vehicle Safety Diagnostic

    3: Signal not available or Invalid
    2: reserved
    1: Safety is OK
    0: Safety is not OK

  - CrashDetnNeg: Crash detection

    3: Info not available (default)
    2: reserved
    1: no crash detected
    0: crash detected

  - SftyInfo_Alvctr: Message Alive Counter(incremented each telegram by 0x01, starts again after 0x0E → 0x00)

- Battery needs every 10 ms “SftyInfo”- message from the vehicle. -

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## VehTi - Message (vehicle time information from vehicle)

- The Battery needs every 1s following vehicle information (descried in ARXML- File):
  - Global time from vehicle necessary

| Name | Start Position | Update Bit | Length [Bit] | Senders | Receivers |
|---|---|---|---|---|---|
| VehTi | 0 | | 64 | Vehicle_Control_... | VIC |
| Year | 0 | | 12 | Vehicle_Control_... | VIC |
| Month | 12 | | 4 | Vehicle_Control_... | VIC |
| Day | 16 | | 6 | Vehicle_Control_... | VIC |
| Hour | 22 | | 5 | Vehicle_Control_... | VIC |
| Minu | 27 | | 6 | Vehicle_Control_... | VIC |
| Sec | 33 | | 6 | Vehicle_Control_... | VIC |

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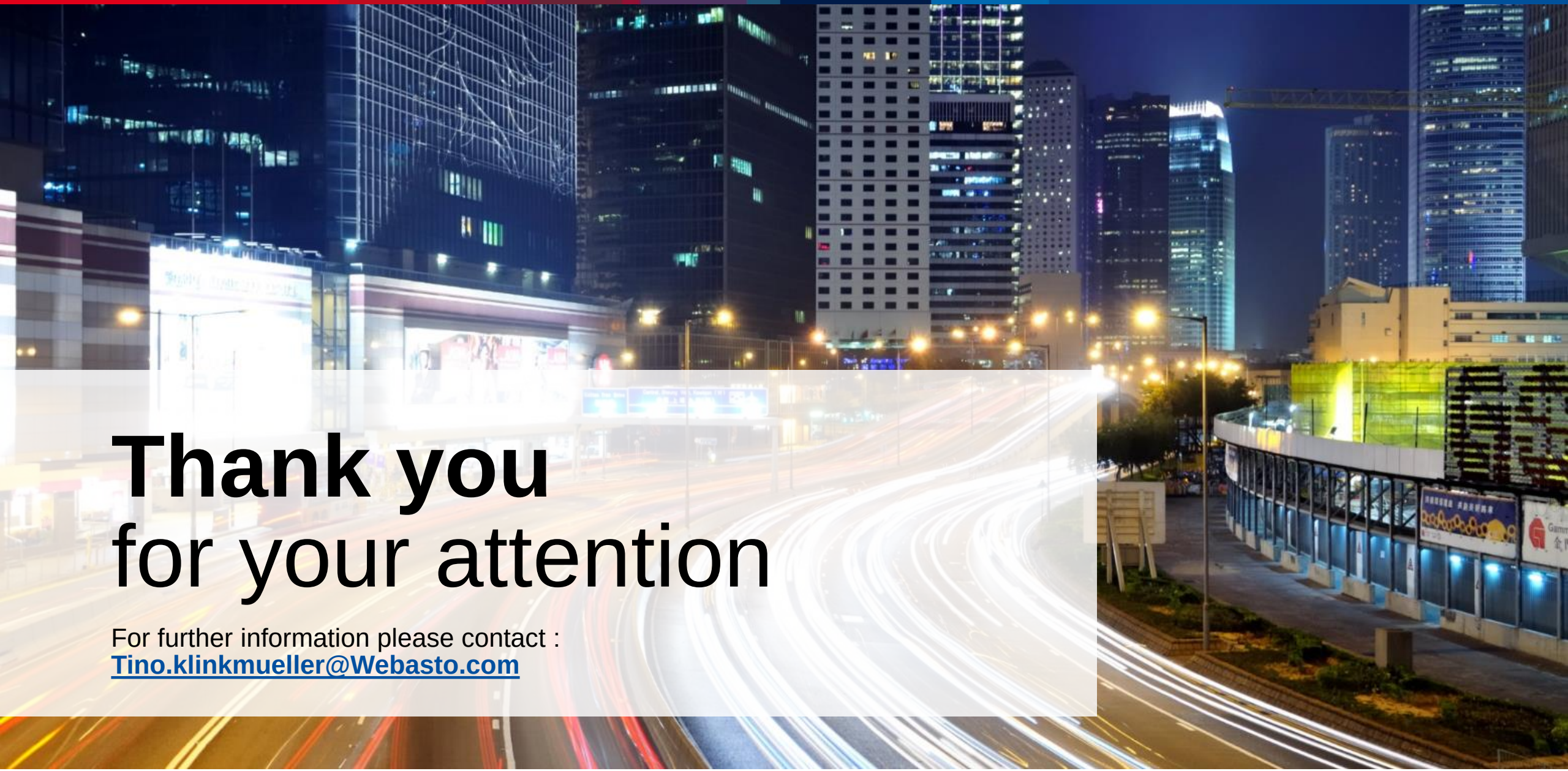
- Battery needs every 1 s “VehTi”- message from the vehicle. -



# Thank you
# for your attention

For further information please contact :
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